

EP 15 Measurement of Acceleration Gravity, Using a Simple Pendulum

1. Objective

To measure the gravitational acceleration (g) using a simple pendulum.

2. Principle

2.1 Core Principle

A simple pendulum (point mass + inextensible, massless string) undergoes simple harmonic motion when displaced by a small angle. The period (T) is given by:

$$T = 2\pi\sqrt{\frac{L}{g}}$$

where L is total pendulum length (string length + bob radius), T is time for one oscillation.

2.2 Linear Regression for Accuracy

To reduce random errors, measure T for multiple L values. Rewrite the period formula as a linear relationship:

$$L = \left(\frac{g}{4\pi^2}\right)T^2 = KT^2$$

where K is slope of the L - T^2 plot, which will be a straight line. Thus $K = \frac{g}{4\pi^2}$. Then g can be obtained by $g = 4\pi^2K$.

3. Procedure

1. **Adjust Initial String Length:** Rotate the screw at the top of the pendulum device to set the string length to approximately 60 cm.
2. **Level the Frame:** Adjust the screws under the device's base to make the frame vertical. Ensure the string is parallel to the frame to avoid conical motion during oscillation.
3. **Measure Bob Diameter:** Measure the diameter of the metal bob (d) 5 times along different directions. Record the zero reading of the vernier caliper (d_0) first, then calculate the average diameter ($\bar{d}$).
4. **Calculate Total Pendulum Length:** Measure the actual length of the string (l) with a meter scale. The total pendulum length is $L = l + \frac{\bar{d}}{2}$ (adding half the bob diameter to account for

the distance from the string's end to the bob's center).

5. Measure Oscillation Time:

- Displace the bob slightly and release it gently to start oscillations.
- Use the stopwatch to record the time for 20 complete oscillations. Repeat this measurement 3 times for the same string length to get consistent data.

6. Repeat for Different Lengths: Adjust the string length to approximately 70 cm, 80 cm, 90 cm, 100 cm, and 110 cm. For each length, repeat Steps 5 to obtain 6 sets of data in total.

4. Data Recording and Processing

4.1 Data Form and Graph

Table 1

Zero reading of the vernier caliper $d_0 =$

Table1 diameter of bob

n	1	2	3	4	5	average
d(m)						

Table 2

Table2

n	Length in m			Time of 20 periods in s				Period T in s	T^2 in s^2
	String l	Radius of bob $d/2$	Total L	1	2	3	mean		
1									
2									
3									
4									
5									
6									

Graph ($L - T^2$)

$K =$, $g =$.

4.2 Data Calculation

4.2.1 Calculate the Gradient (K)

Use linear regression and two-point method to find the gradient of the straight line. And $K =$

4.2.2 Compute Gravitational Acceleration (g)

Using $g = 4\pi^2 K$ (where $\pi \approx 3.1416$):

$g =$

4.2.3 Uncertainty Estimation

- **Length uncertainty:** $u(L)$
- **Time uncertainty:** $u(\bar{t}_{20})$
- **Total uncertainty of g :** $u(g)$

4.2.4 Final Expected Result

$g =$

4.3 Data Analysis

4.3.1 Data Validity Verification

4.3.2 Quality of Linear Fitting

4.3.3 Result Consistency Analysis

4.3.4 Error Source Validation

5. Error Analysis

5.1 Systematic Errors

- **String Extensibility:** String elongation increases L and T , lowering g slightly.
- **Parallax Error in Length Measurement:** Misalignment between the eye and meter scale may overestimate L , contributing to decrease in g .
- **Air Resistance:** As noted, drag increases T , leading to decrease in g .

5.2 Random Errors

- **Human Reaction Time:** Averaging 5 trials reduces this error for t_{30} .
- **Angle Fluctuation:** Small deviations from 5° ($\pm 0.5^\circ$) cause negligible changes in T (< 0.01 s), contributing to uncertainty in g .

6. Questions

1. Why does the small-angle approximation make the pendulum's motion simple harmonious motion?

For small angle, $\sin \theta$ can approximate as θ , which linearizes the restoring force. This linear force satisfies the simple harmonious motion condition ($F = -kx$); if not, θ deviates from $\sin \theta$, and T depends on θ .

2. Why is averaging multiple trials t_{30} more reliable than measuring a single oscillation time (t_1)?

Measuring t_1 amplifies reaction time error. Averaging 5 trials of t_{30} reduces the relative uncertainty.